
Curriculum Vitae of Ilyas Fatkhullin

Website: fatkhullin.github.io Email: fatkhullin.ilyas@gmail.com

Positions

Stanford University, USA 10/2026 – Present
Postdoctoral Scholar, Department of Management Science and Engineering
Mentors: Prof. Renyuan Xu and Prof. Madeleine Udell

Education

ETH Zurich, Switzerland 12/2021 – 09/2026
Ph.D. in Computer Science, ETH AI Center Fellow
Advisor: Prof. Niao He
Research visit: **Georgia Institute of Technology**, USA (with Prof. Guanghai Lan), 03–09/2025
Dissertation Title: Structure, Geometry, and Noise in Stochastic First-Order Optimization

Technical University of Munich, Germany 09/2020 – 11/2021
M.Sc. in Mathematics
External Advisor: Prof. Peter Richtárik Local Advisor: Prof. Michael Ulbrich
Thesis Topic: Error Compensation Method for Communication Efficient Distributed Training

Moscow Institute of Physics and Technology, Russia 09/2016 – 08/2020
Advisor: Prof. Boris Polyak
Thesis Topic: Optimization Landscape of Linear Quadratic Regulator Problem with Output Feedback

Publications ([Google Scholar](#))

Refereed Journal Publications

- [1] I. Fatkhullin, F. Hübler, G. Lan. *Can SGD Handle Heavy-Tailed Noise?*
Accepted to SIAM Journal on Optimization, 2026. Preprint: [arXiv:2508.04860](https://arxiv.org/abs/2508.04860). 2025.
 - [Oral Presentation Award](#) at NeurIPS’25 Workshop on “Optimization for Machine Learning”.
- [2] I. Fatkhullin, N. He, Y. Hu. *Stochastic Optimization under Hidden Convexity*.
[SIAM Journal on Optimization](#), 2025.
- [3] J. Wu, A. Barakat, I. Fatkhullin, N. He. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last-Iterate Convergence*.
[SIAM Journal on Control and Optimization](#), 2025 (preliminary version at [CDC](#), 2023).
- [4] I. Fatkhullin, I. Sokolov, E. Gorbunov, Z. Li, P. Richtárik. *EF21 with Bells & Whistles: Six Algorithmic Extensions of Modern Error Feedback*.
[Journal of Machine Learning Research](#), 2025.
- [5] I. Fatkhullin, B. Polyak. *Optimizing Static Linear Feedback: Gradient Method*.
[SIAM Journal on Control and Optimization](#), 2021.
 - [Cited over 150 times](#).
- [6] B. Polyak., I. Fatkhullin. *Use of Projective Coordinate Descent in the Fekete Problem*.
[Computational Mathematics and Mathematical Physics](#), 2020.

Under Review in Journals

- [1] C. Tsai, I. Fatkhullin, L. Zhang, N. He. *Lower Bounds for Anytime Acceleration of Gradient Descent*.
Preprint: [arXiv:2607.02053](https://arxiv.org/abs/2607.02053), 2026.
- [2] I. Fatkhullin, N. He, G. Lan, F. Wolf. *Global Solutions to Non-Convex Functional Constrained Problems with Hidden Convexity*. Preprint: [arXiv:2511.10626](https://arxiv.org/abs/2511.10626), 2025.
 - [Oral Presentation Award](#) at NeurIPS’25 Workshop “Constrained Optimization for Machine Learning”.

Refereed Machine Learning Conference Publications

- [1] F. Wolf, I. Fatkhullin, N. He. *Global Optimality for Constrained Exploration via Penalty Regularization*. [NeurIPS](#), 2026.
 - [Oral Presentation Award](#) at the European Workshop on Reinforcement Learning (EWRL'26).
- [2] A. Sadiev, P. Richtárik, I. Fatkhullin. *Second-order Optimization under Heavy-Tailed Noise: Hessian Clipping and Sample Complexity Limits*. [NeurIPS](#), 2025.
- [3] F. Sun, I. Fatkhullin, N. He. *Natural Gradient VI: Guarantees for Non-Conjugate Models*. [NeurIPS](#), 2025.
- [4] R. Islamov, Y. As, I. Fatkhullin. *Safe-EF: Error Feedback for Nonsmooth Constrained Optimization*. [ICML](#), 2025.
- [5] F. Hübler*, I. Fatkhullin*, N. He. *From Gradient Clipping to Normalization for Heavy-Tailed SGD*. [AISTATS](#), 2025. *Equal contribution.
- [6] I. Fatkhullin, N. He. *Taming Nonconvex Stochastic Mirror Descent with General Bregman Divergence*. [AISTATS](#), 2024.
- [7] I. Fatkhullin, A. Tyurin, P. Richtárik. *Momentum Provably Improves Error Feedback!* [NeurIPS](#), 2023.
- [8] J. Yang, X. Li, I. Fatkhullin, N. He. *Two Sides of One Coin: the Limits of Untuned SGD and the Power of Adaptive Methods*. [NeurIPS](#), 2023.
- [9] A. Barakat, I. Fatkhullin, N. He. *Reinforcement Learning with General Utilities: Simpler Variance Reduction and Large State-Action Space*. [ICML](#), 2023.
- [10] I. Fatkhullin, A. Barakat, A. Kireeva, N. He. *Stochastic Policy Gradient Methods: Improved Sample Complexity for Fisher-non-degenerate Policies*. [ICML](#), 2023.
- [11] P. Richtárik, I. Sokolov, I. Fatkhullin, E. Gasanov, Z. Li, E. Gorbunov. *3PC: Three Point Compressors for Communication-Efficient Distributed Training and a Better Theory for Lazy Aggregation*. [ICML](#), 2022.
 - [Spotlight Presentation Award at ICML Main Track](#) (top 5% submissions).
- [12] I. Fatkhullin*, J. Etesami*, N. He, N. Kiyavash. *Sharp Analysis of Stochastic Optimization under Global Kurdyka-Lojasiewicz Inequality*. [NeurIPS](#), 2022. *Equal contribution.
- [13] P. Richtárik, I. Sokolov, I. Fatkhullin. *EF21: A New, Simpler, Theoretically Better, and Practically Faster Error Feedback*. [NeurIPS](#), 2021.
 - [Oral Presentation Award at NeurIPS Main Track](#) (top 1% submissions).
 - [Cited over 300 times](#).

Fellowships and Awards

Swiss National Science Foundation (SNSF) Postdoc Fellowship, awarded Project: <i>Structure Exploiting Optimization for Data-Efficient Reinforcement Learning and Games</i> Host: Stanford University	06/2026
Gold Reviewer Award by ICML'2026	05/2026
President's Postdoctoral Fellowship Program, Georgia Institute of Technology Competitive fellowship offer, declined in favor of SNSF	04/2026
Selected for Rising Stars Workshop in Industrial Engineering IOE-ISyE-MS&E Event jointly organized by the UMichigan, Georgia Tech and Stanford	03/2026
Oral Presentation Award from OPT 2025 Workshop at NeurIPS'25	11/2025
Oral Presentation Award from COML 2025 Workshop at NeurIPS'25 for the <i>Best fundamental (theoretical or algorithmic) contribution</i>	10/2025
TMLR Expert Reviewer	09/2025

Rising Star in AI Award, KAUST	02/2023
ICCOPT 2022 Travel Grant	05/2022
Spotlight Presentation Award at ICML	07/2022
Oral Presentation Award at NeurIPS	12/2021
ETH AI Center Doctoral Fellowship	02/2021
German Academic Exchange Service (DAAD) Scholarship for MSc in Germany	04/2020
PreDoc Program Fellowship in Mathematics, Technical University of Munich	03/2020

Teaching Experience

Lecturer, “Data Analysis and Machine Learning”, Ashesi University, Ghana 05/2025
Part of the [Ashesi–ETH Master in Mechatronic Engineering](#).

- Independently designed and delivered a 3-week intensive graduate-level course, including syllabus, lectures, coding labs, assignments, a final project, and exam.
- Student evaluations (26/34 response rate): overall instructor rating 4.45/5.00.
- Received particularly strong scores in preparation of lessons (4.54), encouraging participation (4.54), accessibility and approachability (4.58), and use of instructional technology (4.50).

Teaching Assistant, “[Optimization for Data Science](#)”, ETH Zurich Spring 2023, 2024

- Led weekly tutorials in a core M.Sc.-level optimization course with 100+ students.
- Co-designed homework assignments and contributed exam questions for midterm and final assessments.
- Held office hours and provided individualized support for students.

Instructor, Seminar “Advanced Topics in Machine Learning”, ETH Zurich Fall 2023, 2024

- Guided research-paper discussions, coached students on presentations and scientific writing.
- Evaluated student presentations and advised on short research exploration projects.

Supervision and Mentoring

Since 2022, I have mentored ten Master’s, semester, and research-project students as a primary supervisor. These collaborations have led to publications in prestigious venues (NeurIPS, ICML, AISTATS, SIAM Journal), and mentees have progressed to Ph.D. programs at ETH Zurich, Caltech, Max Planck Institute.

Ph.D. Project Supervision

- Rustem Islamov (University of Basel) — Ph.D. project.
- Yarden As (ETH Zurich) — Ph.D. project.

Master’s Theses

- Chung-En Tsai (ETH Zurich, 2026) — subsequently Ph.D. student, ETH Zurich.
- Fangyuan Sun (ETH Zurich, 2026) — subsequently Quant Researcher, WorldQuant (China).
- Florian Wolf (TU Darmstadt, 2025) — subsequently Ph.D. student, Caltech.
- Jiduan Wu (ETH Zurich, 2023) — subsequently Ph.D. student, Max Planck ETH CLS.
- Harish Rajagopal (ETH Zurich, 2022) — subsequently Software Developer, PQFORCE (Switzerland).

Research Assistants and Interns

- Florian Hübler (ETH Zurich, 2024) — subsequently Ph.D. student, ETH Zurich.

Semester Projects

- Lucas Whitfield (ETH Zurich, 2026) — continued to Master’s Thesis under my supervision at ETH Zurich.

- Fangyuan Sun (ETH Zurich, 2025) — continued to Master’s Thesis under my supervision at ETH Zurich.

Research Internships and Short-term Visits

Amazon , Luxembourg Applied Scientist Intern in Forecasting & Optimization Team Worked on large-scale capacity management problems of Amazon’s delivery network.	10/2025-02/2026
Georgia Institute of Technology , Atlanta, USA H. Milton Stewart School of Industrial and Systems Engineering with Prof. Guanghui (George) Lan Studied properties of stochastic gradient descent under infinite variance.	03–09/2025
King Abdullah University of Science and Technology , Saudi Arabia (remote) with Prof. Peter Richtárik Worked on federated learning algorithms with momentum and quantization.	03–09/2021
École Polytechnique Fédérale de Lausanne , Switzerland (remote) with Prof. Sebastian Stich and Prof. Martin Jaggi Studied accelerated methods for convex optimization.	06–10/2020
German Electron Synchrotron (DESY) , Hamburg; with Prof. Judith Katzy Applied adversarial learning methods to the detection of events in high-energy physics.	07–09/2019
Helmholtz-Zentrum Berlin , Germany; with Prof. Ji Li Developed numerical solvers for physical simulations in materials science.	07–08/2018

Professional Activities

Journal Reviewer: Mathematical Programming; Operations Research; SIAM Journal on Optimization; SIAM Journal on Control and Optimization; Journal of Machine Learning Research; IEEE Transactions on Automatic Control; Transactions on Machine Learning Research.

Conference Reviewer: ICML (2022-2025); AISTATS 2023; NeurIPS (2023-2026); ICLR 2024.

Workshop Organizer: [Swiss Optimization Symposium](#), Monte Verità Ascona, Switzerland 08/2026

Session Organizer at SIAM Optimization Conference 06/2026

Location: Edinburgh, United Kingdom

Session Title: “Optimization under Data-driven Constraints”

Session Organizer at INFORMS Optimization Society Conference 03/2026

Location: Atlanta, USA

Session Title: “Methods for Multi-stage Decision Making”

Session Organizer at International Conference on Continuous Optimization 07/2025

Location: University of Southern California, Los Angeles, USA

Cluster and Session Title: “Optimization for Data Science”; “Adaptive Methods in Optimization”

Session Organizer at INFORMS Optimization Society Conference 07/2024

Location: Rice University, Houston, USA

Cluster and Session Title: “Optimization in Data Science”; “Recent Advances in Min-Max Optimization”

Selected Talks

I. Fatkhullin. *Global Optimization under Hidden Convexity: Non-convex Functional Constraints* 06/2026
SIAM Optimization Conference, University of Edinburgh, United Kingdom

I. Fatkhullin. *Can SGD Handle Heavy-Tailed Noise?* 05/2026
Lund University, Lund, Sweden

I. Fatkhullin. <i>Natural Gradient Variational Inference: Guarantees for Non-conjugate Models</i> . SlidesLive NeurIPS 2025 , San Diego, USA	11/2025
I. Fatkhullin. <i>Can SGD Handle Heavy-Tailed Noise?</i> H. Milton Stewart School of ISyE, Georgia Institute of Technology, Atlanta, USA	08/2025
I. Fatkhullin. <i>Safe Error Feedback with Applications in Humanoid Robot Fleet Training</i> . International Conference on Stochastic Programming (ICSP 2025), Paris, France	08/2025
I. Fatkhullin. <i>Nonconvex Stochastic Mirror Descent with Implications in Machine Learning</i> . International Conference on Continuous Optimization (ICCOPT 2025), Los Angeles, USA	07/2025
I. Fatkhullin. <i>Heavy-Tailed Gradients in AI Models</i> . ETH AI Center Seminar, Zurich, Switzerland	12/2024
I. Fatkhullin. <i>Stochastic Optimization under Hidden Convexity</i> . European Conference on Advances in Continuous Optimization (EUROPT 2024) , Lund, Sweden	07/2024
I. Fatkhullin. <i>Momentum Provably Improves Error Feedback!</i> SlidesLive NeurIPS 2023 , New Orleans, USA	12/2023
I. Fatkhullin. <i>Stochastic Policy Gradient Methods for Fisher-non-degenerate Policies</i> . SlidesLive ICML 2023 , Honolulu, USA	07/2023
I. Fatkhullin. <i>Policy Gradient Methods in Reinforcement Learning</i> . Rising Star in AI Symposium, KAUST, Saudi Arabia	02/2023
I. Fatkhullin. <i>Stochastic Optimization under Kurdyka-Lojasiewicz Condition</i> . Institute of Machine Learning Seminar, Zurich, Switzerland	10/2022
I. Fatkhullin. <i>Stochastic Optimization under Kurdyka-Lojasiewicz Condition</i> . ETH AI Center Doctoral Seminar, Zurich, Switzerland	06/2022
I. Fatkhullin. <i>Practical Algorithmic Extensions of Modern Error Feedback</i> . Federated Learning One World (FLOW) Seminar	10/2021

References

Prof. Niao He

Associate Professor, Department of Computer Science
ETH Zurich, Switzerland
Email: niao.he@inf.ethz.ch

Prof. Peter Richtárik

Professor of Computer Science
King Abdullah University of Science and Technology (KAUST), Saudi Arabia
Email: peter.richtarik@kaust.edu.sa

Prof. Guanghui (George) Lan

A. Russell Chandler III Chair and Professor, ISyE
Georgia Institute of Technology, Atlanta, USA
Email: george.lan@isye.gatech.edu